Objectives

- ✓ Classes in Java.

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Class

- ✓ A class defines a new data type.
- ✓ Class is the template for an object.
- ✓ A class is declared by use of the class keyword.
- ✓ Contains - Instance variable & member functions(methods).

Class Definition

```
class classname {
   type instance variable1;
   type instance variable2;
   //..
   type instance variable;
   type methodname1(parameter-list){
        //body of the method
   }
   type methodname1(parameter-list){
        //body of the method
   }
}
```

A Simple Class

```
class Box
{
        double width;
        double height;
        double depth;
}
```

- ✓ Box is the new type.
- ✓ Here, class declaration creates a template only.

Object Creation

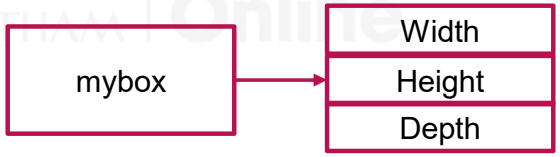
Box mybox = new Box(); // create a Box object called mybox

- ✓ Every **Box** object will contain its own copies of the instance variables **width**, **height**, and **depth**.
- ✓ To access these variables, you will use the *dot* (.) operator. mybox.width = 100;
- ✓ This statement will assign width 100 to mybox object.

Object Representation

Box mybox;

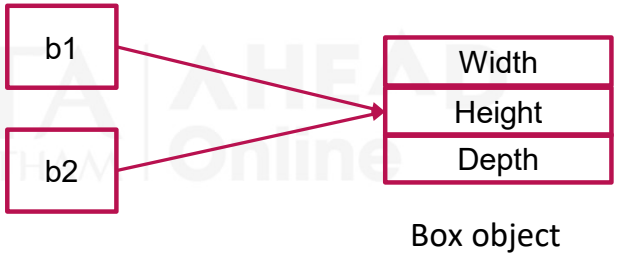
mybox=new Box();



Object Assignment

Box b1 = new Box();

Box b2 = b1;



Methods

General form of a method

```
type name(parameter-list)
        {
                // body of method
        }
```

- ✓ type is the return type.
  - ✓ If the method does not return a value, its return type must be **void**.
- ✓ The name of the method is specified by *name*.

Example

```
// This program includes a method inside the box class.
class Box {
        double width;
        double height;
        double depth;
        // display volume of a box
        void volume() {
                System.out.print("Volume is ");
                System.out.println(width * height * depth);
        }
}
```

Example

```
class BoxDemo3
{
        public static void main(String args[]) {
          Box mybox1 = new Box();
          Box mybox2 = new Box();
          // assign values to mybox1's instance variables
          mybox1.width = 10;
          mybox1.height = 20;
          mybox1.depth = 15;

        // assign different values to mybox2's instance variables
          mybox2.width = 3;
          mybox2.height = 6;
          mybox2.depth = 9;

          mybox1.volume(); // display volume of first box
          mybox2.volume(); // display volume of second box

        }
}
```

Takeaways

- ✓ To define a class.
- ✓ Object creation and internal representation.
- ✓ Example.

Reference

[1] Herbert Schildt, *"Java: The Complete Reference, Eleventh Edition", Oracle 2018*